

Duration-Aware Phone Embedding Upsampling for Open Vocabulary Keyword Spotting

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Introduction

- Task of identifying the instances of a query in continuous speech.

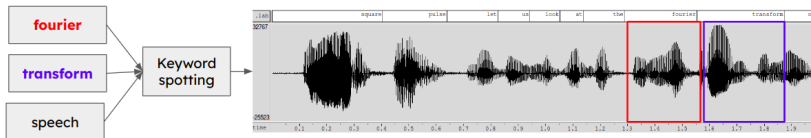


Figure: Identifying the keywords in given speech

- Often used in
 - Telephone tapping
 - Voice activated devices
 - Information retrieval from Spoken document archives.

- Several end-to-end models have been proposed for closed vocabulary.
- Projecting the speech signal and keyword into common latent space and making the decision
- Use of fixed-length embeddings, leading to the loss of temporal information and an inability to localize keywords.
- ASR-free KWS system¹ predicts the presence of the keyword within the given audio.
- End-to-end open-vocabulary keyword spotting² method predicts keyword endpoints using CTC.

¹Audhkhasi, et al. End-to-end ASR-free keyword search from speech. ICASSP (2017).

²Wei, Bo et al. End-to-End Transformer-Based Open-Vocabulary Keyword Spotting with Location-Guided Local Attention. Interspeech (2021).

Motivation & Problem Statement

- Proposed lattice-free KWS method that:
 - Doesn't require lattice extraction
 - Doesn't require joint training of ASR with CTC
 - Solves the out-of-vocabulary issue

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- Given a keyword $\mathbf{Q} = (q_1, q_2, \dots, q_K)$ and acoustic features $\mathbf{X} = (x_1, x_2, \dots, x_N)$, seek $Y_{loc} = (y_1, y_2, \dots, y_N)$ and $Y_{det} = \{0, 1\}$.

$$Y_{loc} = \begin{cases} 1, & \text{if the keyword } q \text{ is spoken within } X \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

Architecture

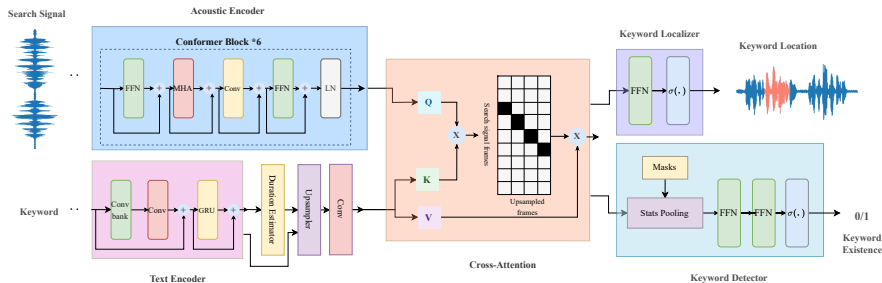


Figure: Architecture of the proposed method.

- Acoustic Encoder

- Takes 128-dimensional mel filter banks of the speech signal
- Conformer Encoder is used as the acoustic encoder to preserve the temporal information and for local and global context modeling.

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- Text Encoder

- Grapheme-to-phoneme converter is used to get phoneme sequence
- CNNs are used to exploit the local information for better capturing the co-articulations of the phonemes.
- CNNs capture contextual information up to K phonemes.
- Highway networks and a bidirectional gated recurrent unit (GRU) are used to capture the long-term dependencies.

Duration Estimator

- The phoneme alignments between audio and text are obtained using Viterbi decoding .
- The embeddings extracted from the text encoder are passed to the duration estimator module to predict each phoneme duration.
- Duration estimator contains 2 feed forward layers with non linear activations.

$$L_{dur} = \frac{1}{M} \sum_{i=1}^M |d_i - \hat{d}_i| \quad (2)$$

- Where d_i is the groundtruth duration for phoneme i and $\hat{d}_i$ is the predicted duration.

- Matches the text phoneme rate to the acoustic frame rate by Upsampling based on the predicted durations.
- To generate unique representations, Sinusoidal positional embeddings are added for every frame of a text keyword.
- During training, Original frame durations are used for upsampling.
- In contrast, during inference, the predicted durations are used to upsample the phoneme embeddings.

Cross attention

- Cross-attention modules generates the cross correlation matrix

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_k}}\right) \quad (3)$$

$$\mathbf{W} = \mathbf{AV} \quad (4)$$

- where $\mathbf{K}$ and $\mathbf{V}$ are extracted from Text embeddings and $\mathbf{Q}$ are extracted from Acoustic embeddings.
- Attention matrix exhibits monotonic pattern if keyword exists otherwise it exhibits obscure pattern.

Loss functions

- Binary cross entropy losses employed at the frame level and utterance level are given by

$$L_{loc} = -\frac{1}{V} \sum_{i=1}^V \sum_{t=1}^T (y_t^i \log(p_t^i) + (1 - y_t^i) \log(1 - p_t^i))$$

$$L_{det} = -\frac{1}{V} \sum_{i=1}^V (y^i \log(p^i) + (1 - y^i) \log(1 - p^i))$$

$$L_{total} = \lambda_1 L_{dur} + \lambda_2 L_{loc} + \lambda_3 L_{det}$$

$y_t^i \rightarrow$ ground truth posterior probability

$p_t^i \rightarrow$ predicted posterior probability

Dataset description

- The proposed method is trained and evaluated on Librispeech Speech corpus
- Training data : 100 hrs of read speech containing 585 speakers.
- Testing : 5 hrs of test clean speech containing 87 speakers
- Evaluated in 2 scenarios, i.e., in-vocabulary and out-of-vocabulary.
- In-vocabulary scenario : keywords seen, audios unseen
- Out-of-vocabulary scenario : keywords unseen, audios unseen

Training details

- Randomly select keywords from sentences containing more than 5 phonemes to create positive examples.
- Randomly select keywords from the lexicon that do not appear in the given utterance to generate negative examples.
- 128-dimensional linear spectrogram is extracted at a frame shift of 10ms and window size of 25ms.
- Initial phone embeddings dimension is 128.
- The model is trained for 100 epochs with an ADAM optimizer with a learning rate of $1e-3$.

- Localization is assessed using the Intersection Over Union (IoU) metric.
- Detection performance is measured using the Mean Average Precision (mAP) score.

Table: Performance evaluation in IV and OOV scenarios

Systems	Metrics				Inference time in sec
	IV		OOV		
	mAP	IOU	mAP	IOU	10sec
W/o duration	61.98	62.01	53.13	58.63	0.056
With duration	71.24	63.33	68.98	66.63	0.066

tSNE visualizations of the phoneme & keyword embeddings

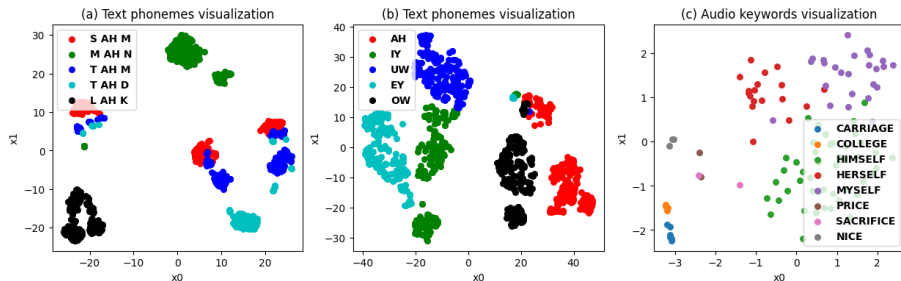


Figure: (a) illustrates the tSNE visualizations of the text /AH/ phoneme occurring in different contexts, (b) illustrates the tSNE visualizations of the vowels (c) illustrates the tSNE visualizations audio keywords

Effect of query length on performance

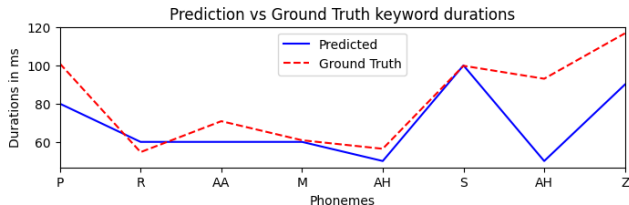


Figure: Predicted and ground truth keyword durations for 'PROMISES'

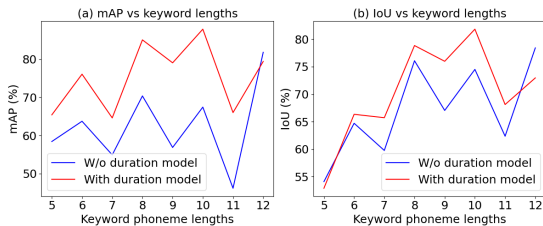


Figure: Effect of query length on performance

Conclusions & Future work

- Proposed a novel strategy to mitigate rate mismatches between audio and text by utilizing a duration estimator and an upsampler.
- Proposed method has robust generalization capabilities in terms of out-of vocabulary .
- Explore the downsampling of the acoustic embeddings to a phoneme rate and perform cross-attention at the phoneme rate

Thank you